

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ 2 SET E

DURATION: 30 MINUTES

SUMMER SEMESTER, 2024-2025

FULL MARKS: 20

CSE 4205: Digital Logic Design

Programmable calculators are not allowed.

Answer **all 2 (two)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

ID:

1. Use Karnaugh map to find the minimal form for the following expression (with don't care terms).
Clearly group the terms contributing to your solution.
- $F(A, B, C, D, E) = \Sigma m(0, 2, 4, 8, 10, 13, 14, 21, 23, 24, 28, 29, 30), d(1, 5, 9, 16, 18, 31)$
- 7
(CO3)
(PO2)

2. Simplify the following expression using Quine McCluskey (Tabular Method). And in the process identify the Prime Implicants and Essential Prime Implicants. 13
(CO5)
(PO3)

$$F(A, B, C, D) = \sum m(0, 5, 8, 11, 13, 15), d(3, 6, 7, 10)$$